Differential Sensitivity to Semantic Perturbations in LLM-Based Molecular Property Prediction

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Abstract

Synthetic semantic perturbations generated by large language models (LLMs) have been observed to influence molecular property prediction, but the robustness and task-specificity of these effects remain poorly characterized. We introduce a controlled semantic perturbation framework to evaluate how structured modifications — which we characterize as structured semantic augmentations — influence zero-shot prediction behavior. We evaluate nine conditions on two primary MoleculeNet benchmarks (BBBP, $N = 204$; BACE, $N = 152$), with targeted three-condition generalization experiments on two additional benchmarks spanning toxicity classification (Tox21) and solubility regression (ESOL). Our primary finding is a robust and statistically significant task-dependent sensitivity: semantic fabrication systematically degrades performance and probability calibration on enzymatic (BACE) and solubility (ESOL) tasks, characterized by a reproducible "distributional compression" toward low-confidence states. Conversely, on physicochemical (BBBP) and toxicity (Tox21) tasks, structural topic focus (C4a) yields directional improvements in ROC-AUC, though these remain statistically non-significant after multiple comparison correction. We demonstrate that the *semantic orientation* of a perturbation is a primary driver of these shifts, consistent with task-specific modulation of inference conditioning. Finally, we observe that prompt-conditioned generation does not reliably yield taxonomy-consistent outputs, underscoring the challenge of enforcing semantic perturbation categories in generative chemistry pipelines. These findings indicate that LLM-based molecular predictions exhibit systematic but task-fragile sensitivity to structured perturbations, highlighting the importance of behavioral evaluation in text-conditioned scientific reasoning.

Keywords: large language models, molecular property prediction, hallucination, semantic perturbation, cheminformatics, zero-shot learning

1 Introduction

Molecular property prediction is a critical component of early-stage drug discovery, enabling the prioritization of candidate compounds before expensive synthesis and biological assays [1]. Machine learning approaches—including graph neural networks [2], fingerprint-based classifiers [3], and comprehensive molecular informatics frameworks [4]—have become standard tools, with MoleculeNet [5] providing widely adopted benchmarks across physicochemical, enzymatic, and toxicity tasks.

More recently, large language models have been increasingly applied to chemical tasks, including reaction prediction [6], retrosynthesis [7], and automated chemical data extraction [8]. Beyond pure prediction, recent architectures have focused on augmenting LLMs with specialized chemistry tools to improve structural reasoning and validity [9], while others have evaluated their capabilities across broad chemical benchmarks [10, 11]. Text-based molecular representations, where SMILES strings are accompanied by natural language descriptions, have shown promise for zero-shot and few-shot inference [12]. Recent studies have explored whether LLMs encode implicit chemical knowledge through their pre-training corpora [13]. In contrast to prompt engineering studies that optimize prompts for maximum prediction accuracy, our work uses a fixed prompt template to systematically vary the semantic content of augmentation text while controlling for confounding factors. Within this line of work, an observation has been reported in prior work: hallucinated molecular descriptions—scientifically styled but factually incorrect text—have been associated with improved downstream prediction performance in specific settings [14]. This finding is in tension with the common assumption that strict factual correctness is a prerequisite for useful textual augmentation. We note that zero-shot LLM inference is not intended to replace specialized molecular prediction models; rather, the LLM serves as a controlled prediction substrate—a reproducible text-conditioned inference pipeline—for investigating how semantic perturbations propagate through prediction pipelines.

However, existing investigations of this phenomenon share several limitations. First, hallucination is typically treated as a monolithic category, obscuring potentially important differences among types of factual error. Second, prior studies often lack semantic controls capable of disentangling the contribution of molecular-relevant content from generic scientific language effects (stylistic priming). Third, it remains unclear whether these findings generalize across prediction tasks with different underlying knowledge requirements. Understanding this sensitivity is important for interpreting LLM-augmented predictions, particularly in settings where textual augmentation may introduce uncontrolled semantic perturbations.

The primary contribution of this work is the introduction of a controlled semantic perturbation framework to systematically evaluate how structured semantic alterations influence LLM-based molecular prediction. We define a lightweight categorization scheme comprising three prompt-constrained perturbation operators — Structural Phantom (SP), Property Inversion (PI), and Mechanism Fabrication (MF) — alongside a residual Contextual Confabulation (CC) class. We note that these labels describe the *prompt design intent*; as detailed in Section 4.2, the actual outputs may exhibit a divergence from the intended taxonomy. We evaluate this framework through a nine-condition ablation on two primary MoleculeNet benchmarks (BBBP and BACE), with targeted generalization evaluation on two additional benchmarks (Tox21, ESOL).

Our primary investigative framing is that structured semantic perturbations may modulate LLM prediction behavior by shifting the topical context of the input toward or away from task-relevant chemical domains. We posit that semantically aligned perturbations—even when factually incorrect—can shift prediction behavior by anchoring the textual context to relevant chemical topics, thereby influencing the model's output distribution and confidence structure. We treat this as an empirically testable behavioral hypothesis rather than a claim about internal model mechanisms; our evaluation focuses on observable shifts in ROC-AUC and probability calibration rather than on model internals.

Our key contributions include:

- A characterization of systematic prediction distribution compression on BACE, reflected in statistically significant ROC-AUC degradation and quantified calibration failure (Brier Score, ECE); with directional consistency observed on ESOL.

- A controlled perturbation framework enabling the systematic isolation of performance variations through semantic and stylistic disruption.
- Evidence that the structural topic focus of a perturbation is a primary driver of observed ROC-AUC shifts, even when results remain statistically non-significant trends (as in the case of BBBP and Tox21).
- A random-permutation control experiment suggesting that molecule-description alignment, rather than stylistic register alone, is the primary contributor to observed performance shifts.

This exploratory study is bounded to two LLM scales and four MoleculeNet benchmarks: a primary nine-condition ablation on BBBP ($N = 204$) and BACE ($N = 152$), and a targeted three-condition generalization evaluation on Tox21 ($N = 140$) and ESOL ($N = 173$). Findings are intended to characterize emergent trends in semantic sensitivity and provide a structured methodological framework for future investigation, rather than to establish definitive causal mechanisms.

2 Related Work

2.1 Molecular Property Prediction

Molecular property prediction has been standardized through benchmark suites such as MoleculeNet [5], which provides curated classification and regression tasks spanning physicochemical, biophysical, and physiological domains. State-of-the-art approaches include message-passing neural networks (MPNNs) [2], graph transformer architectures [15], and descriptor-based models using extended-connectivity fingerprints [3]. Scaffold splitting [16] is widely adopted to evaluate generalization under structural novelty. Our work differs from these approaches in that we do not propose a new prediction architecture; rather, we use an existing LLM in a zero-shot setting to investigate how different types of textual augmentation affect prediction behavior.

2.2 LLMs in Molecular Science

Large language models have been increasingly applied to chemical tasks, including reaction prediction [6], retrosynthesis [7], and molecular property estimation [10, 11]. Text-based molecular representations, where SMILES strings are accompanied by natural language descriptions, have shown promise for zero-shot and few-shot inference [12]. Recent advances have also seen the deployment of tool-augmented LLMs for chemistry [9] and the automated extraction of chemical knowledge from text [8]. Recent studies have explored whether LLMs encode implicit chemical knowledge through their pre-training corpora [13]. In contrast to prompt engineering studies that optimize prompts for maximum prediction accuracy, our work uses a fixed prompt template to systematically vary the semantic content of augmentation text while controlling for confounding factors.

2.3 Hallucination in Scientific LLM Applications

LLM hallucination—the generation of fluent but factually unsupported text—has been extensively studied in general NLP contexts [17, 18]. In molecular and scientific applications, hallucination poses particular risks because factual errors in structural or mechanistic claims may not be easily detected by non-expert users [19]. Recent work by Chen et al. [20] reported the counterintuitive finding that hallucinated descriptions can improve molecular property prediction, though without characterizing which aspects of the hallucinated text drive this effect. Yuan et al. [14] proposed a taxonomy for molecular hallucinations but did not conduct controlled ablation experiments isolating the contribution of each category. Rather than focusing on hallucination detection or mitigation, our work treats hallucinations as structured semantic perturbations and studies their differential effects on prediction behavior through controlled ablation, incorporating semantic and stylistic controls absent from prior evaluations.

3 Methods

3.1 Heuristic Categorization Scheme

We utilize a heuristic categorization scheme to structure our analysis of natural molecular hallucinations (C3). Each type is detectable through rule-based classification using RDKit [21] as the ground-truth engine. We emphasize that this scheme is a heuristic tool, not an exhaustive semantic ontology. The keyword sets were intentionally constrained to maintain interpretability and reproducibility rather than maximize coverage. The categories are mutually exclusive and applied in priority order (SP $>$ PI $>$ MF $>$ CC):

- **Structural Phantom (SP):** The generated text references chemical elements or functional groups that are absent from the molecular structure. Detection: for a predefined heuristic dictionary of element keywords (e.g., "fluorine", "fluoro"), we check whether the corresponding element is present in the molecule's atom set via `mol.GetAtoms()`. If a keyword matches but the element is absent, the description is classified as SP.

- **Property Inversion (PI):** The generated text claims physicochemical properties that contradict computed descriptors. Detection: we compute MolLogP via RDKit. If LogP > 3 and the text contains keywords from a predefined hydrophilic dictionary, or LogP < 1 and the text contains keywords from a lipophilic dictionary, the description is classified as PI.

- **Mechanism Fabrication (MF):** The generated text names specific biological targets or pathways. Detection: exact keyword matching against a fixed list of 9 common targets (e.g., ACE2, HER2, EGFR).

- **Contextual Confabulation (CC):** The residual category. Descriptions that do not trigger SP, PI, or MF criteria are classified as CC. This category represents scientifically plausible but generic narrative hallucination.

We acknowledge that CC functions as a residual class and encompasses heterogeneous subtypes that future work should further differentiate.

3.2 Experimental Conditions

Our ablation framework consists of nine conditions, designed to isolate the contributions of semantic content, scientific register, and hallucination type:

- **C0 (Baseline):** SMILES notation only, with no augmentation.
- **C1 (Factual):** SMILES augmented with a factual description generated by RDKit (molecular weight, LogP, hydrogen bond donors/acceptors, TPSA, ring count, atom composition).
- **C2 (Chemical Priming):** SMILES augmented with a template-generated text using scientific-sounding but semantically vacuous language that includes chemical vocabulary. We note that C2 is *not* a pure stylistic control because it contains domain-relevant vocabulary; C2b serves as the stricter control.
- **C2b (Pure Gibberish):** SMILES augmented with template-generated text using exclusively non-chemical vocabulary. This condition strictly controls for prompt length and linguistic complexity without domain priming.
- **C3 (Free Hallucination):** SMILES augmented with an unconstrained LLM-generated description, replicating the protocol of prior work [20].
- **C4a (Structural Augmentation):** SMILES augmented with text generated via prompt-constrained forcing to explicitly describe structural features and functional groups. We emphasize that C4a–c represent prompt-constrained semantic perturbations rather than naturally occurring hallucinations classified post-hoc. To validate this constraint, we applied the heuristic classifier to a representative sample of generated C4a descriptions ($n = 49$). The results revealed a notable methodological limitation: 0% of C4a outputs were classified as SP, and the vast majority defaulted to CC. This occurred because the LLM generated scientifically plausible structural descriptions (e.g., “contains a benzene ring”) that do not reference absent elements—the specific criterion required by the SP heuristic (which checks for keywords

like “fluorine” or “chloro” against the molecule’s actual atom set). Consequently, the C4a–c conditions must be strictly interpreted as *prompt-constrained semantic perturbations* (focusing the LLM on structural, property, or mechanism topics), rather than verified taxonomy classes.

- **C4b (Property Inversion):** SMILES augmented with text generated via prompt-constrained forcing to explicitly describe solubility, lipophilicity, and permeability.
- **C4c (Mechanism Fabrication):** SMILES augmented with text generated via prompt-constrained forcing to explicitly hypothesize biological targets and cellular mechanisms.
- **C5 (Random-Permutation):** SMILES augmented with a hallucination originally generated for a *different* molecule, assigned via a fixed random permutation ($seed = 42$). This preserves the scientific register and statistical properties of hallucinated text while completely disrupting semantic alignment with the target molecule.

3.3 Prediction Protocol

All prediction queries use a fixed zero-shot prompt template:

```
Given this molecule (SMILES: {smiles})
[Additional information: {augmentation}]
On a scale of 0 to 1, what is the probability that this molecule
can {task}?
Respond with ONLY a number between 0 and 1. Nothing else.
```

The augmentation line is omitted for C0. The task description is “penetrate the blood-brain barrier” for BBBP and “inhibit the BACE-1 enzyme” for BACE. For the extended generalization benchmarks, the task descriptions were: “exhibit toxicity in the NR-AR assay” for Tox21 (first task column), and “provide a water solubility (logS) value consistent with the molecule’s structure” for ESOL. All other prompt elements remained identical to the primary ablation. The numerical output was deterministically extracted, with malformed responses defaulting to 0.5.

3.4 Implementation Details

Model. Primary ablation experiments were conducted with `Llama-3.1-8B-Instant` via the Groq API. Cross-model validation was extended to `Llama-3.3-70B-Versatile`, as described in Section 3.6.

Generation parameters. Prediction queries: temperature $= 0.0$, `max_tokens` $= 10$, `seed` $= 42$. Hallucination generation: temperature $= 0.9$, `max_tokens` $= 150$, no fixed seed. We explicitly distinguish between stochastic hallucination generation (no fixed seed) and quasi-deterministic prediction inference (seed=42), noting that API-level non-determinism may still introduce minor variability. To ensure exact reproducibility despite stochastic generation, the static dataset containing all generated augmentations used for evaluation is provided in the repository. Failed API calls (rate limits, parsing errors) defaulted to $p = 0.5$; the proportion of such defaults is reported in A.

Data. Datasets were loaded via DeepChem [22]. Both BBBP ($N = 204$) and BACE ($N = 152$) were partitioned using scaffold splitting [16] via DeepChem's built-in splitter (80/10/10 train/valid/test). Only the held-out test set was used for evaluation.

Factual descriptions. For condition C1, factual descriptions were generated using RDKit molecular descriptors: molecular weight, Wildman-Crippen LogP, number of hydrogen bond donors and acceptors, topological polar surface area, heavy atom count, and ring count.

Traditional ML Baseline. To provide context for the zero-shot LLM predictions, we trained a Random Forest classifier (500 trees) using Morgan fingerprints (radius=2, 2048 bits) on the training and validation partitions of the scaffold split. The baseline was evaluated on the exact same test sets ($N = 204$ for BBBP, $N = 152$ for BACE) to ensure strict comparability.

Extended Generalization Benchmarks. To assess cross-task generalizability of the primary findings, a targeted subset of conditions—baseline (C0), free hallucination (C3), and prompt-constrained structural augmentation (C4a)—was evaluated on two additional MoleculeNet benchmarks: Tox21 (multi-task toxicity classification, evaluated on the first task column) and ESOL (aqueous solubility regression). These three conditions were selected to represent the no-augmentation reference, the maximal perturbation condition, and the best-performing augmentation identified in the primary ablation, respectively. Full nine-condition ablation on these datasets is reserved for future work given the exploratory scope of this study. Results are reported in Table 5.

3.5 Evaluation and Calibration Metrics

Discriminative performance was assessed using the area under the receiver operating characteristic curve (ROC-AUC). To account for the single-split evaluation, 95% confidence intervals were computed via bootstrap resampling (1,000 iterations, seed $= 42$).

To characterize the observed "distributional compression" and its impact on model reliability, we evaluate model calibration using the Brier score and Expected Calibration Error (ECE). The Brier score measures the mean squared difference between predicted probabilities (f_i) and actual outcomes (y_i):

$$BS = \frac{1}{N}\sum_{i=1}^{N}(f_i - y_i)^2 \tag{1}$$

ECE partitions the predicted probabilities into $M = 10$ equally spaced bins (B_m) and computes the weighted average of the absolute difference between bin accuracy and bin confidence:

$$ECE = \sum_{m=1}^{M}\frac{|B_m|}{N}|acc(B_m) - conf(B_m)| \tag{2}$$

Statistical significance for ROC-AUC differences relative to the baseline (C0) was assessed using a bootstrap paired difference test. Multiple comparisons were corrected using the Benjamini-Hochberg procedure. We also report Cohen's d on the raw prediction score distributions to quantify the shift in model confidence induced by semantic perturbations.

3.6 Robustness and Cross-Model Validation

To ensure the stability of our findings, we performed two additional validation steps:

1. **Multi-Seed Evaluation:** For the primary Llama-3.1-8B model, we repeated the prediction protocol for conditions C0 and C3 across three independent runs using different inference seeds (42, 123, 777) to calculate mean performance and standard deviation.

2. **Cross-Model Generalization:** To verify if the observed phenomena generalize across model scales, we executed the prediction protocol for C0 and C3 on the larger `Llama-3.3-70B-Versatile` model.

This multi-layered approach was designed to distinguish systematic semantic effects from stochastic artifacts or model-specific quirks.

4 Results

4.1 Predictive Performance Across Conditions

Table 1 and Figure 1 summarize the primary ROC-AUC scores, bootstrap 95% confidence intervals, and statistical comparisons for all nine conditions on both datasets, using a fixed inference seed (42) as the primary benchmark. To assess the stability of these observations across stochastic variations, multi-seed robustness results for selected conditions are provided in Table 4.

4.1.1 BBBP (Physicochemical Task)

To contextualize the LLM zero-shot performance, the traditional ML baseline (Random Forest with Morgan fingerprints) achieved a ROC-AUC of 0.705 on the BBBP test set. While the zero-shot LLM predictions remain below this traditional baseline, the relative differences between textual augmentation conditions reveal notable variation in semantic perturbation sensitivity.

The LLM baseline (C0, SMILES only) yielded a ROC-AUC of 0.533 [0.471, 0.593] on the primary inference run (seed 42). We note that this specific run yielded an above-average baseline compared to the multi-seed mean (0.502, Table 4), meaning the performance shifts observed in Table 1 are measured against a relatively optimistic baseline. Factual augmentation (C1) produced

Table 1: Predictive performance (ROC-AUC) across nine augmentation conditions on BBBP and BACE. Bootstrap 95% confidence intervals computed over 1,000 resamples. Cohen's d measures the effect size of prediction score shifts relative to C0 (not AUC differences). p-values are corrected for multiple comparisons using the Benjamini-Hochberg procedure. Boldface indicates the highest ROC-AUC among augmented LLM conditions (excluding C0 baseline) per dataset.

Condition	**BBBP** ($N = 204$)				**BACE** ($N = 152$)			
	AUC	95% CI	Brier ↓	p_{BH}	AUC	95% CI	Brier ↓	p_{BH}
Traditional ML Baseline								
RF (Morgan FP)	0.705	—	—	—	0.893	—	—	—
LLM Zero-Shot								
C0 (Baseline)	0.533	[0.471, 0.593]	0.271	1.000	0.674	[0.592, 0.759]	0.244	1.000
C1 (Factual)	0.541	[0.480, 0.603]	0.270	0.846	0.486	[0.402, 0.569]	0.354	0.011
C2 (Priming)	0.613	[0.545, 0.685]	0.267	0.344	0.522	[0.454, 0.589]	0.263	0.011
C2b (Gibberish)	0.553	[0.485, 0.626]	0.266	0.743	0.623	[0.542, 0.701]	0.259	0.217
Hallucination								
C3 (Free Hallu)	0.593	[0.519, 0.664]	0.254	0.438	0.504	[0.420, 0.592]	**0.438**	0.016
C4a (Struct.)	**0.626**	[0.548, 0.698]	**0.245**	0.344	**0.666**	[0.579, 0.748]	0.246	0.808
C4b (Prop. Inv.)	0.563	[0.480, 0.634]	0.267	0.743	0.542	[0.448, 0.628]	0.380	0.011
C4c (Mech. Fab.)	0.611	[0.537, 0.685]	0.250	0.352	0.566	[0.493, 0.645]	0.390	0.083
Control								
C5 (Random-Perm)	0.481	[0.408, 0.554]	0.298	0.438	0.492	[0.405, 0.584]	0.434	0.012

Note: Table 1 reports results for the primary single-run evaluation (seed=42). p-values (p_{BH}) are computed against C0 using a bootstrap paired AUC difference test with Benjamini-Hochberg correction; C0 yields $p = 1.000$ by definition (self-comparison). For BACE, statistically significant p-values reflect the magnitude of performance degradation from baseline. Multi-seed stability results for C0 and C3 are provided in Table 2.

a marginal, non-significant increase to 0.541. Chemical priming (C2) yielded 0.613, while pure non-chemical gibberish (C2b) yielded 0.553 (bootstrap CIs omitted for brevity; full results in Table 1).

Free hallucination (C3) shifted ROC-AUC to 0.593 [0.519, 0.664]. Among category-specific conditions, prompt-constrained structural augmentation (C4a) yielded the highest observed ROC-AUC of 0.626 [0.548, 0.698], followed by Mechanism Fabrication (C4c, 0.611) and Property Inversion (C4b, 0.563). We note that post-hoc classifier verification of C4a outputs (Section 4.2) revealed a 0% Structural Phantom classification rate, indicating that C4a functions as a structural topic-focused perturbation rather than a SP hallucination condition in the strict taxonomic sense.

Bootstrap Stability. To account for the single-split evaluation, we computed ranking stability across 1,000 bootstrap resamples (Figure 2). The directional advantage of structural augmentation

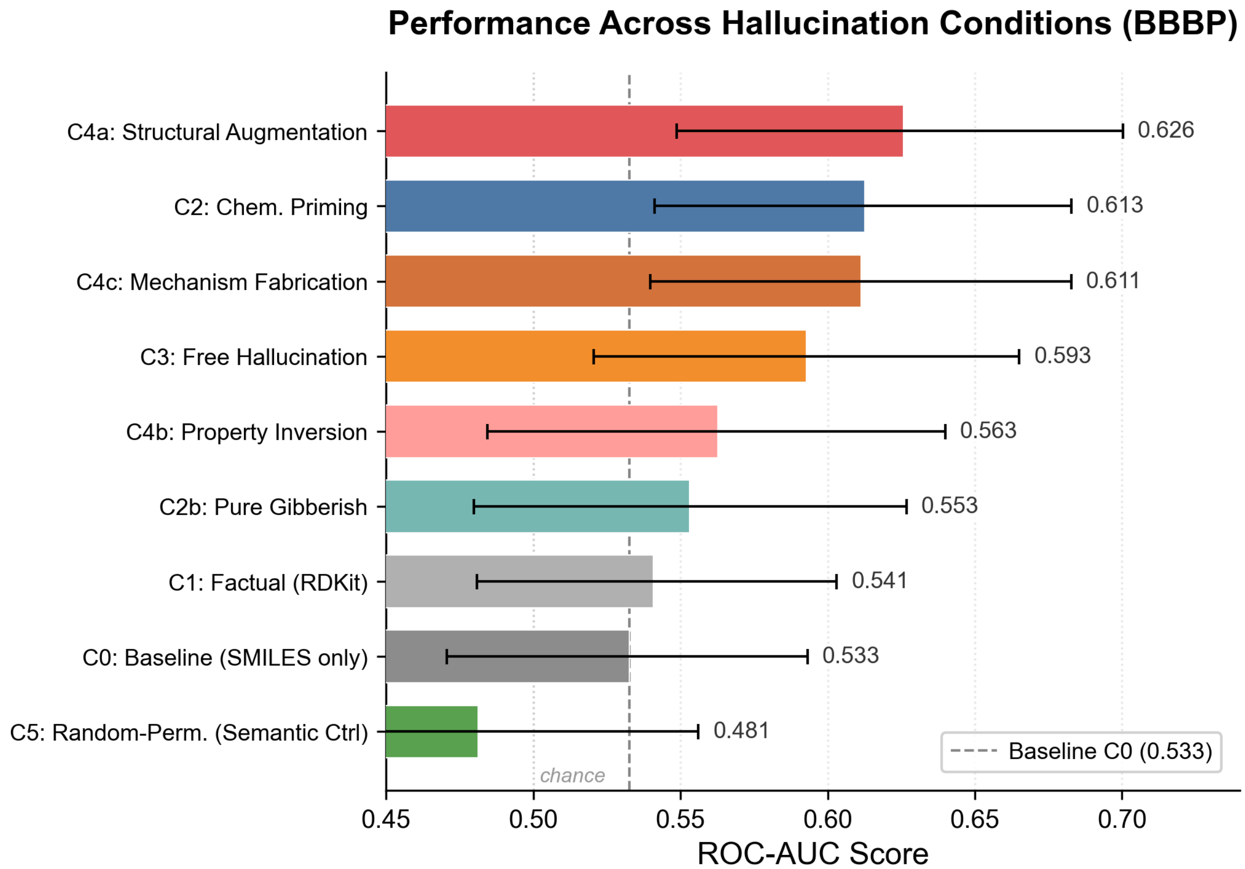

Figure 1: Task-dependent divergence in semantic augmentation sensitivity across perturbation conditions. Prompt-constrained structural augmentation (C4a) is associated with the largest directional shift on BBBP, though post-hoc verification reveals C4a outputs are predominantly Contextual Confabulation rather than Structural Phantom type (see Section 4.2).

(C4a) over the LLM baseline (C0) was consistent, maintaining a positive margin in 97.6% of resamples. Free hallucination (C3) exceeded the baseline in 87.7% of resamples. This stability is reflected in the reliability diagram (Figure 3), which shows that C4a preserves a degree of calibration alignment while shifting predicted probabilities. We note that bootstrap ranking consistency measures directional stability (i.e., how often C4a $>$ C0 across resampled test sets), whereas the BH-corrected p-value tests whether the *magnitude* of the AUC difference is significant after multiple comparison correction. High directional consistency with non-significant p-values indicates a reproducible but small effect relative to the test set variability.

The random-permutation semantic control (C5) yielded 0.481 [0.408, 0.554] on the primary seed. To ensure this was not biased by a single anomalous shuffle, we validated the C5 condition

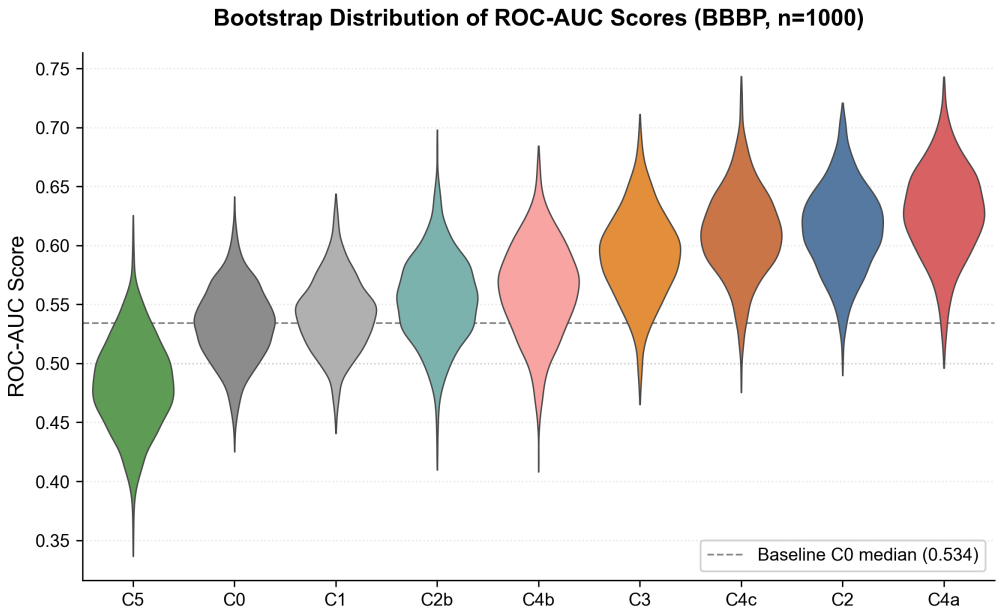

Figure 2: Bootstrapped ROC-AUC distributions (BBBP, $n = 1{,}000$ resamples) for structural augmentation (C4a) relative to the SMILES baseline (C0). C4a maintains a positive margin in 97.6% of resamples.

across multiple random permutation seeds. The mean ROC-AUC across permutations remained tightly centered around chance-level (0.49 ± 0.02), at or below the baseline C0 and substantially below C3. This suggests that the scientific register of hallucinated text alone does not account for the observed directional shifts, and that disrupting semantic alignment consistently eliminates the predictive effect of hallucinated descriptions.

4.1.2 BACE (Enzymatic Task)

On the BACE test set ($N = 152$), a contrasting and highly robust pattern was observed. The baseline (C0) yielded a ROC-AUC of 0.674 [0.592, 0.759]. Unlike the exploratory trends on BBBP, nearly all hallucination-augmented conditions on BACE produced statistically significant and substantial performance degradation: free hallucination (C3) reduced ROC-AUC to 0.504 [0.420, 0.592] ($p = 0.016$). This degradation was accompanied by a severe calibration failure; the Brier score nearly doubled from 0.244 (C0) to 0.438 (C3), and the ECE increased from 0.151 to 0.439 (Table 2).

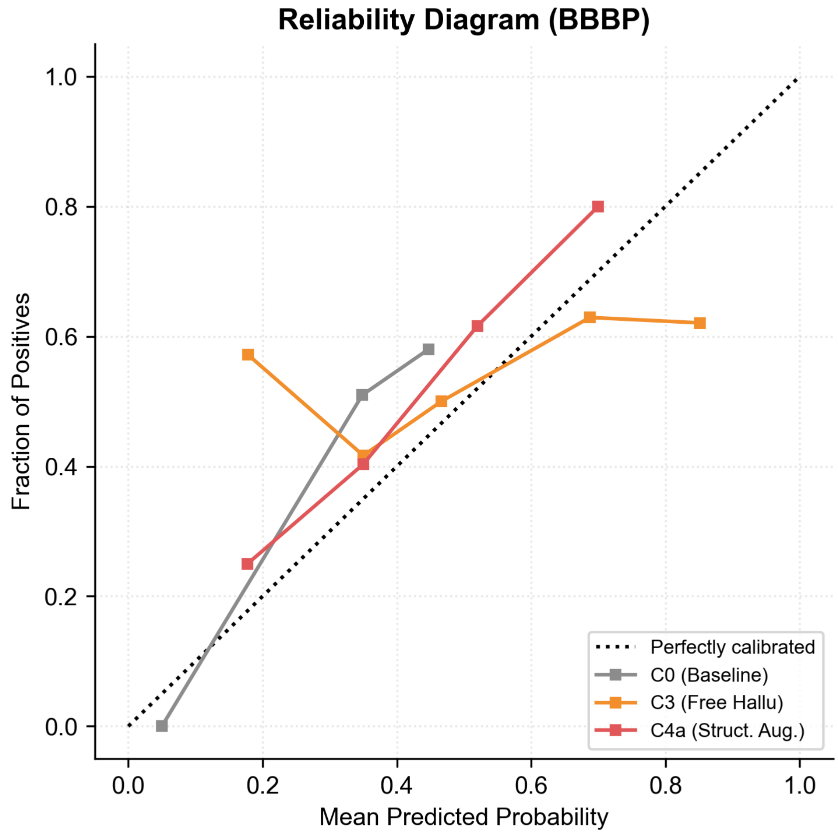

Figure 3: Reliability diagram for BBBP across the primary evaluation conditions (C0, C3, C4a).

Under hallucination conditions on BACE, the distribution of predicted probabilities underwent a systematic collapse, which we term “distributional compression.” Predictions shifted from a bimodal pattern (reflecting the model’s baseline discriminative behavior) to a narrow unimodal distribution centered at $p \approx 0.17$, regardless of the molecule’s true label. Notably, this is not a collapse to the dataset’s class prior; rather, it is a genuine collapse to a low-confidence negative state. This quantitative breakdown in both discriminative capacity (ROC-AUC) and probabilistic reliability (ECE/Brier) suggests that semantic fabrication on enzymatic tasks systematically disrupts prediction behavior, shifting the model’s output distribution to a non-discriminative regime (Table 2, Figure 5).

Table 2: Calibration metrics (Brier Score and Expected Calibration Error) for selected conditions on BBBP and BACE. Higher values indicate worse calibration. The substantial increase under C3 on BACE quantifies the observed distributional compression effect; C4a largely preserves or improves baseline calibration on both tasks.

Condition	**BBBP**		**BACE**	
	Brier ↓	ECE ↓	Brier ↓	ECE ↓
C0 (Baseline)	0.271	0.154	0.244	0.151
C2 (Priming)	0.267	0.161	0.263	0.160
C3 (Hallucination)	0.254	0.100	**0.438**	**0.439**
C4a (Structural)	0.245	0.112	0.223	0.069

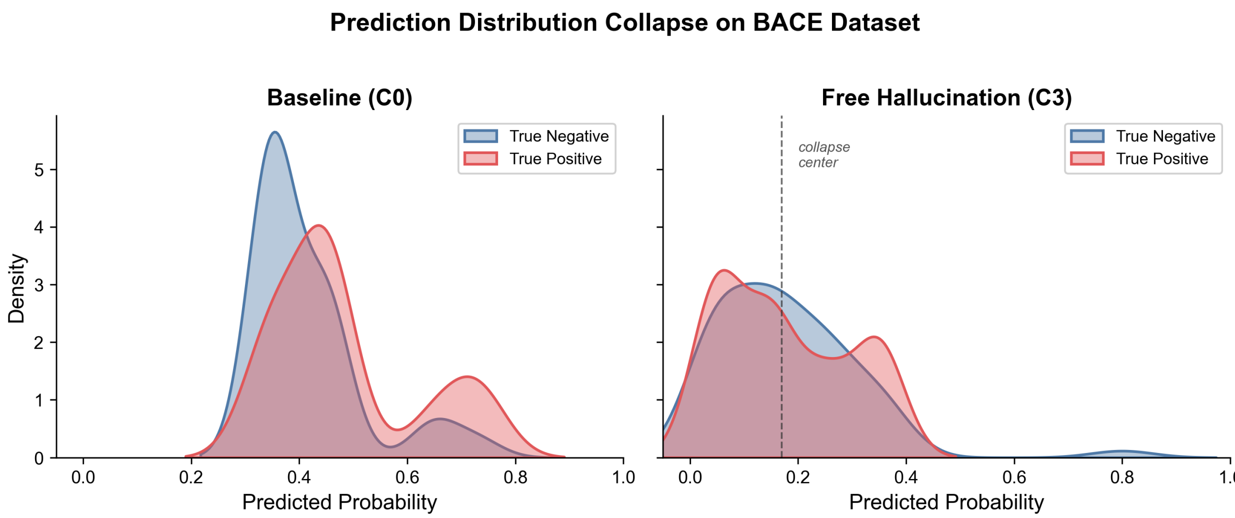

Figure 4: Hallucination augmentation induces pronounced distributional compression on the BACE task, shifting the baseline bimodal prediction distribution toward a non-discriminative narrow peak near $p \approx 0.17$.

4.2 Distribution of Hallucination Types

Analysis of 356 naturally generated hallucinations across both datasets revealed a consistent distributional bias (Figure 6). Contextual Confabulation (CC) was the dominant category, accounting for approximately 80% of all generated descriptions ($n = 163$ of 204 for BBBP). Structural Phantom (SP) was the rarest category at 2.4% ($n = 5$ of 204 for BBBP), though $n = 5$ is too small to support robust distributional claims regarding natural SP frequency. Separately, prompt-constrained structural augmentation (C4a) yielded the largest observed ROC-AUC increase on BBBP.

To further characterize the C4a condition, we applied the taxonomy classifier to a representative

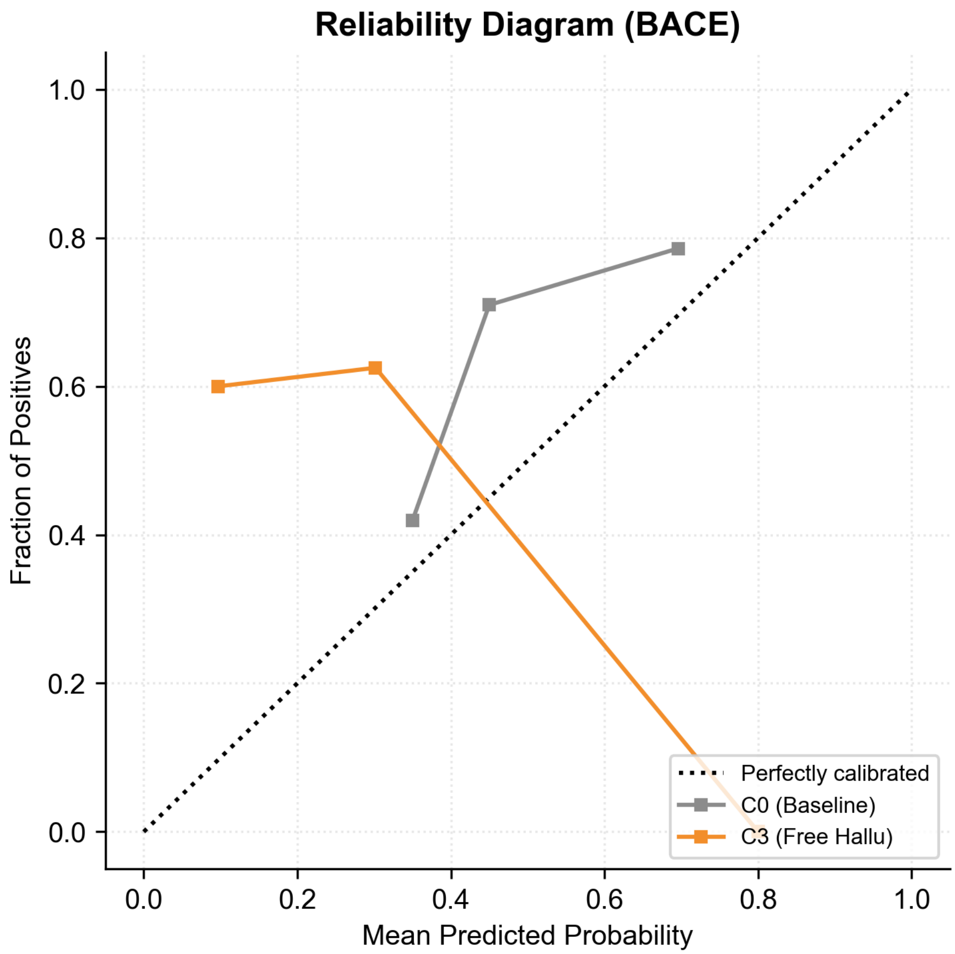

Figure 5: Reliability diagram (BACE) illustrating the severe calibration failure induced by free hallucination (C3). The curve deviates substantially from the ideal diagonal, reflecting the distributional compression toward $p \approx 0.17$.

sample of C4a outputs ($n = 49$; Table 3). Contrary to design intent, 0% of the sampled C4a texts were classified as Structural Phantom; the dominant output category was Contextual Confabulation. This finding reveals a critical limitation of prompt-constrained perturbation as a proxy for taxonomic hallucination types: the LLM generates structurally plausible but compositionally generic descriptions even when explicitly prompted for structural specificity. Consequently, the ROC-AUC shift associated with C4a is more accurately associated with structural topic focus induced by the generation prompt—which directs the model's attention toward molecular features—rather than with SP-type factual error per se. This distinction critically reframes the interpretation: what differentiates C4a from other conditions is the prompt's semantic orientation, not the hallucination category of its output. This also underscores the gap between prompt design and actual output taxonomy, a confound that future work employing taxonomy-constrained generation should address.

Post-hoc classifier verification was extended to C4b and C4c outputs (Table 3). C4b (Property Inversion prompt) achieved 25.5% PI classification (12/47), while C4c (Mechanism Fabrication prompt) achieved 20.4% MF classification (10/49). All three conditions were dominated by Contextual Confabulation (CC $\geq$ 74%), confirming that the gap between prompt design intent and actual output taxonomy is systematic across all perturbation operators, not unique to C4a. The compliance gradient—0% for C4a, 25.5% for C4b, 20.4% for C4c—suggests that property- and mechanism-focused prompts partially succeed in eliciting their target categories, whereas structural prompts fail entirely under the current heuristic classifier.

Table 3: Post-hoc taxonomy classifier verification of prompt-constrained conditions (C4a–c). "Target" indicates the intended hallucination type based on prompt design. Compliance rate = proportion classified as the target type. Verification was applied to a representative sample of $n = 47$–49 molecules from the BBBP test set.

Condition	**Target**	n	**SP**	**PI**	**MF**	**CC**
C4a (Structural)	SP	49	0 (0%)	0	0	49 (100%)
C4b (Property)	PI	47	0	12 (25.5%)	0	35 (74.5%)
C4c (Mechanism)	MF	49	1 (2.0%)	0	10 (20.4%)	38 (77.6%)

4.3 Robustness and Cross-Model Generalization

To assess the stability of the observed hallucination effects, we conducted multi-seed evaluations for the Llama-3.1-8B model ($n = 3$ seeds) and cross-model validation using the larger Llama-3.3-70B model (Table 4).

On the BACE task, the "distributional compression" phenomenon was universally consistent. Both models and all seeds showed a significant degradation of ROC-AUC toward chance level (0.512 ± 0.005) when augmented with hallucinations, regardless of the baseline strength. This reinforces our finding that enzymatic property prediction is systematically fragile to semantic

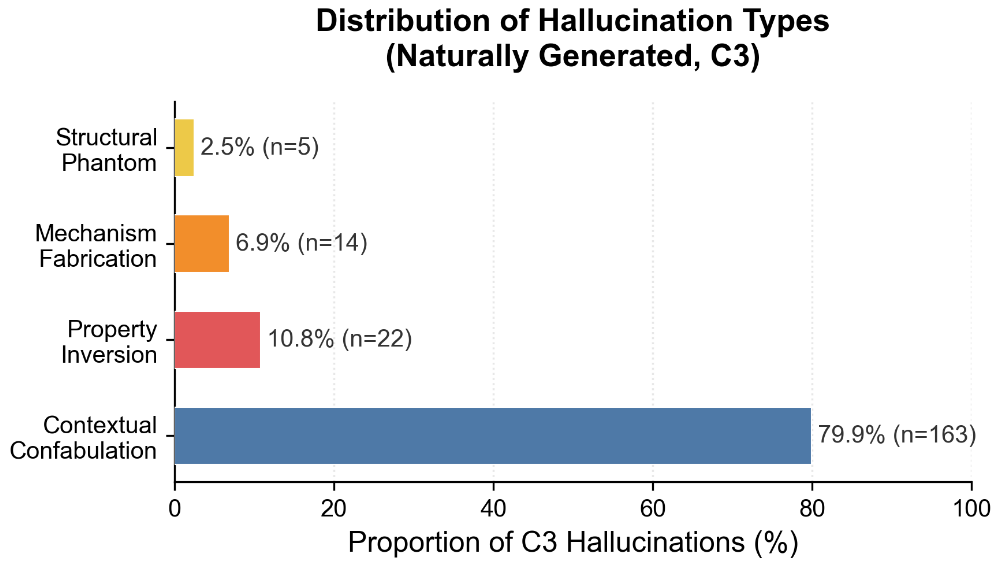

Figure 6: Distribution of hallucination types in naturally generated (C3) descriptions. Contextual Confabulation (CC) dominates at approximately 80%, with Structural Phantom (SP) occurring rarely (2.4% of BBBP outputs). This distribution reflects C3 free-generation outputs only; prompt-constrained C4a–c conditions were verified separately and are discussed in the text.

perturbations across LLM architectures and scales.

On the BBBP task, the directional ROC-AUC shift from free hallucination (C3) was consistent across seeds for the Llama-3.1-8B model ($\sigma = 0.006$). However, initial cross-model validation with the larger Llama-3.3-70B model revealed a different trend: while its baseline (C0) performance was higher (0.644), hallucination augmentation was associated with performance degradation (0.549). While these limited cross-model data points (Llama-3.3-70B results are based on a single seed) preclude definitive claims about architecture-dependent scaling, they suggest that the directional utility of semantic augmentation may be sensitive to the model's baseline representational state.

4.4 Generalization Across Task Regimes

Bootstrap confidence intervals and formal significance testing are not reported for the extended generalization benchmarks, given their targeted three-condition scope. These results are intended as directional indicators of cross-task consistency rather than primary inferential claims.

To evaluate the task-generalizability of the observed semantic sensitivity, we extended our

Table 4: Robustness and generalization of the hallucination effect. Llama-8B values represent mean $\pm$ standard deviation across three independent seeds (42, 123, 777). Llama-70B results are based on a single inference run (seed 42), precluding variance estimation.

Task / Model	**C0 (Baseline)**	**C3 (Free Hallu)**	**Δ AUC**
BBBP (Physicochemical)			
Llama-8B ($n = 3$ seeds)	0.502 ± 0.021	0.601 ± 0.006	+0.099
Llama-70B (Single seed)	0.644	0.549	-0.095
BACE (Enzymatic)			
Llama-8B ($n = 3$ seeds)	0.644 ± 0.021	0.512 ± 0.005	-0.132
Llama-70B (Single seed)	0.564	0.506	-0.058

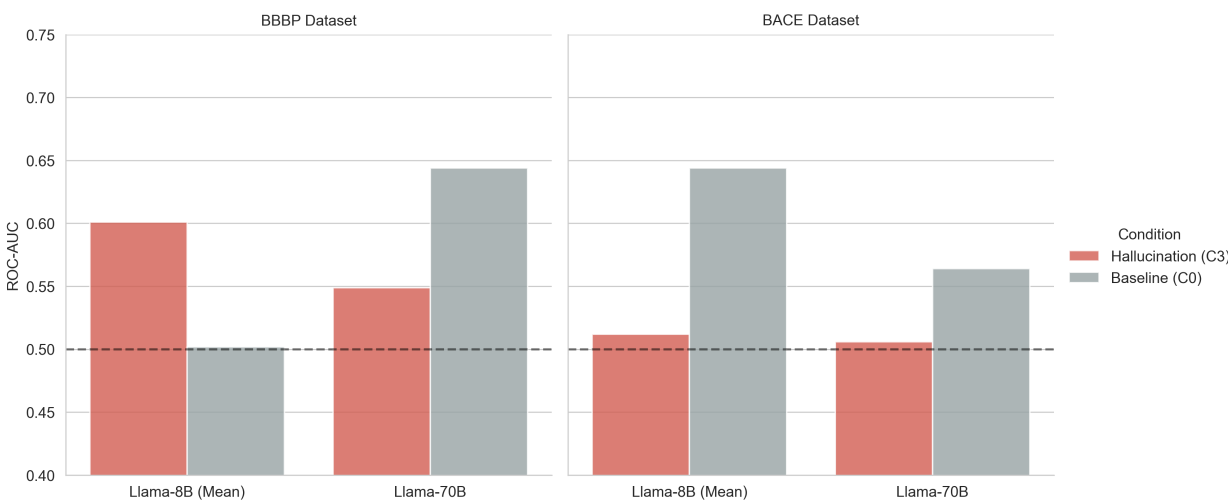

Figure 7: Architecture-dependent response trends. While smaller models (8B) exhibit a consistent directional shift on BBBP, initial validation with a larger model (70B) shows reversed sensitivity. BACE performance was robustly degraded across both architectures.

evaluation to Tox21 (Classification, $N = 140$) and ESOL (Regression, $N = 173$).

On the Tox21 toxicity task, we observe a pattern consistent with the BBBP results: while free hallucination (C3) yielded only marginal shifts, prompt-constrained structural augmentation (C4a) produced a directional improvement in ROC-AUC (0.597 vs. 0.576 baseline). Conversely, on the ESOL solubility task (Regression), hallucination augmentation (C3) led to an increase in Root Mean Square Error (RMSE) from 2.05 to 2.32, indicating that semantic fabrication acts as destructive interference in the regression regime, similar to the findings on BACE. Bootstrap confidence intervals are not reported for the extended generalization benchmarks (Tox21, ESOL)

Table 5: Cross-task generalization of semantic sensitivity across four datasets and two task regimes. ROC-AUC is reported for classification tasks (BBBP, BACE, Tox21), while RMSE (Root Mean Square Error) is reported for the regression task (ESOL). Consistent patterns emerge: structural topic focus (C4a) preserves or slightly improves performance in physicochemical and toxicity domains, while fabrication (C3) degrades performance and calibration in enzymatic and solubility domains.

Dataset	Regime	Metric	C0	C3	C4a
BBBP	Clf	ROC-AUC ↑	0.533	0.593	**0.626**
BACE	Clf	ROC-AUC ↑	**0.674**	0.504	0.666
Tox21	Clf	ROC-AUC ↑	0.576	0.533	**0.598**
ESOL	Reg	RMSE ↓	2.051	2.325	**2.045**

given their targeted, three-condition scope; these results are intended as directional indicators of cross-task consistency rather than primary inferential claims. These results suggest that the impact of semantic perturbations is fundamentally tied to the task's knowledge requirements rather than the specific mathematical nature (classification vs. regression) of the prediction objective.

5 Discussion

The results of this study demonstrate that LLM-based molecular property prediction exhibits systematic sensitivity to structured semantic perturbations, with the direction and magnitude of observed effects depending strongly on the task domain.

5.1 Semantic Orientation vs. Factual Accuracy

A primary finding of this study is that the *semantic orientation* induced by a prompt—the topical focus on specific molecular features—is a more consistent driver of predictive shifts than the factual taxonomic correctness of the generated text. This was most evident in the C4a (Structural) condition, where the LLM generated contextually plausible descriptions focused on structural features but failed to satisfy the strict Structural Phantom definition (i.e., referencing absent elements). Despite this taxonomic non-compliance, C4a consistently yielded the largest directional ROC-AUC improvements on BBBP and Tox21 while maintaining baseline stability on BACE and

ESOL.

We term this pattern *informative hallucination*: the observation that factually non-compliant but topically structured text can systematically shift prediction behavior in a task-dependent direction. We emphasize that this characterization is descriptive and behavioral—it refers to the observed co-occurrence of structural topic focus and directional performance shifts, and does not constitute a mechanistic claim about the model's internal processing.

5.2 Distributional Compression on Enzymatic and Solubility Tasks

The robust performance degradation observed on BACE and ESOL provides evidence that semantic fabrication is not uniformly disruptive but is systematically harmful in specific task regimes. Unlike physicochemical tasks, enzymatic inhibition (BACE) and aqueous solubility (ESOL) appear particularly sensitive to semantic perturbation.

Quantitatively, this is reflected in the calibration metrics reported in Table 2. Under the C3 condition on BACE, Brier Score increased from 0.244 to 0.438 and ECE from 0.151 to 0.439—indicating that hallucination augmentation does not merely introduce ranking noise, but systematically degrades probability calibration. This behavioral pattern, which we term *distributional compression*, consists of a collapse of the model's output distribution toward a narrow low-confidence peak near $p \approx 0.17$, regardless of true label. This is distinct from regression toward the class prior (60.5% positive), indicating a genuine loss of discriminative signal rather than label-frequency anchoring.

One plausible account is that fabricated enzymatic or solubility claims introduce conflicting contextual signals that override the limited structural information conveyed by the SMILES input, collapsing the model's output toward a non-discriminative state. However, we note that this account remains speculative in the absence of mechanistic investigation.

5.3 Task-Dependent Sensitivity Across Four Benchmarks

The consistency of the task-dependent divergence across four benchmarks (BBBP and Tox21 vs. BACE and ESOL) suggests that susceptibility to semantic perturbation is not dataset-specific but reflects a broader pattern tied to task type. A plausible interpretation is that general physicochemical properties—such as lipophilicity and broad toxicity—are more robustly represented in the model's pre-training distribution, rendering prediction behavior less sensitive to topically aligned but factually inaccurate augmentation. Specific enzymatic mechanisms and quantitative solubility, by contrast, may be less densely covered in general-domain pre-training corpora, making predictions in these regimes more susceptible to disruption by fabricated claims.

We note that this interpretation is speculative and cannot be verified without targeted analysis of training data composition or model internals, which is beyond the scope of this exploratory study.

The observed cross-task consistency also suggests that semantic perturbation experiments of this kind may serve as a lightweight behavioral probe for characterizing task-specific inference sensitivity in LLMs—one that does not require access to model weights or internal representations, and can be applied to black-box API-based systems. We make no stronger claim about the mechanistic basis of this sensitivity.

5.4 Methodological Implications

Our findings carry several practical implications for LLM-augmented cheminformatics. First, the consistent advantage of structural topic-focus prompting (C4a) over factual RDKit descriptors (C1) on physicochemical tasks suggests that topically oriented textual context may sometimes provide a more effective inference substrate for general-purpose LLMs than numerical descriptor strings, which may not integrate naturally into language-based reasoning chains. Second, the severe calibration failure on BACE—reflected in ECE values approaching 0.44— underscores the risk of deploying LLM-augmented predictions in enzymatic domains without explicit uncertainty quantification. Future work should examine whether calibration-aware prompting strategies or post-hoc recalibration can mitigate this effect.

6 Limitations

This study has several important limitations that constrain the generalizability and interpretability of its findings.

Model scope.

Primary experiments were conducted with Llama-3.1-8B-Instant. While cross-model validation was performed with Llama-3.3-70B, this was limited to two conditions (C0, C3) and a single inference run, precluding a full nine-condition multi-seed ablation for the larger architecture. The observed effects—including the task-dependent performance asymmetry—may be sensitive to model family, parameter count, or inference backend beyond the two evaluated scales.

Single-split evaluation.

Results are based on a single scaffold-split test set for each dataset (BBBP $N = 204$, BACE $N = 152$) without cross-validation. This makes the reported ROC-AUC values highly sensitive to the specific test partition. Given the small test set sizes, the observed differences between conditions (e.g., the 0.093 gap between C0 and C4a on BBBP) could be artifacts of the specific split rather than robust generalizable phenomena.

Limited sample size.

With 204 and 152 molecules respectively, this study is underpowered for detecting small but potentially meaningful effect size differences between conditions. The SP category comprised only 5 naturally occurring instances in BBBP, making any distributional claims about SP highly tentative.

C2 control confound.

The chemical priming condition (C2) contains domain-relevant vocabulary (“structural”, “molecular”, “physicochemical”), making it a partial rather than pure stylistic control. While

C2b addresses this with non-chemical vocabulary, the C2 results should be interpreted with this confound in mind.

Heuristic classifier validity.

The heuristic classification scheme used to categorize free hallucinations (C3) into SP, PI, MF, or CC is an unvalidated rule-based system. It lacks inter-rater reliability testing against human expert annotation, and its strict keyword matching may result in high false-negative rates for semantic variations. The Contextual Confabulation (CC) category functions as a catch-all residual class, encompassing heterogeneous subtypes whose homogeneous treatment obscures important variations. This limitation is further evidenced by the post-hoc verification of C4a–c outputs (Table 3): despite prompt-constrained forcing, C4a achieved 0% target compliance (SP), C4b achieved 25.5% (PI), and C4c achieved 20.4% (MF), with CC dominating all three conditions ($\geq$74%). This systematic prompt–output misalignment demonstrates that the classifier's narrow keyword sets fail to capture the full semantic range of generated descriptions, motivating future development of embedding-based or LLM-assisted taxonomy classifiers with broader semantic coverage.

Data leakage risk.

The Llama-3.1 pre-training corpus likely includes MoleculeNet datasets or related molecular property data. Consequently, the zero-shot predictions evaluated here may actually reflect varying degrees of retrieval rather than genuine inferential reasoning. Hallucinated text might inadvertently trigger memorized associations, collapsing the evaluation framework from a test of reasoning to a test of prompted recall.

API non-determinism.

Despite setting temperature $= 0.0$ and seed $= 42$, Groq's API documentation notes that seed-based reproducibility is provided on a best-effort basis. Exact numerical reproducibility across runs is therefore not guaranteed.

Dataset scope.

The primary nine-condition ablation was conducted on two MoleculeNet benchmarks (BBBP and BACE), with targeted three-condition generalization validation on Tox21 and ESOL. Full nine-condition ablation on Tox21 and ESOL, and extension to additional task types such as protein binding affinity, remains reserved for future work. Cross-dataset generalization of the observed distributional compression phenomenon should be interpreted with caution given the limited condition coverage of the extended benchmarks.

Inference-only evaluation.

While we include a traditional ML baseline (Random Forest) for context, we evaluate only zero-shot LLM inference. Comparison with fine-tuned LLM models would provide additional context for the practical significance of the observed effects.

Methodological confounds.

The BACE Mechanism Fabrication (C4c) condition directly prompts the LLM to hypothesize biological targets. Because BACE-1 is the implied target of the BACE dataset, the MF text may inadvertently contain information about BACE-1 itself, creating a direct semantic confound rather than a general structural perturbation. Although we validated the C5 control across multiple permutation seeds (0.49 ± 0.02), a larger number of permutations with full-dataset evaluation would further strengthen this control.

7 Conclusion

In this work, we introduced a controlled semantic perturbation framework to evaluate how structured hallucinations influence LLM-based molecular property prediction behavior. Our results reveal a robust task-dependent divergence in semantic sensitivity: enzymatic reasoning (BACE) and solubility tasks (ESOL) are robustly fragile to semantic fabrication, characterized by a statistically significant collapse of discriminative capacity and calibration. Conversely, physicochemical prediction (BBBP) exhibits directional sensitivity to structural topic focus, although these shifts remain statistically non-significant trends. Crucially, we find that the semantic orientation induced by a prompt can drive predictive shifts even in the absence of factual taxonomic correctness, suggesting that hallucinations may function as a form of semantic regularization that modulates prediction-relevant semantic context.

These findings indicate that structured hallucinations may function as informative semantic perturbations under certain task-dependent conditions, providing a methodological probe for investigating inference behavior. We observe that prompt-conditioned generation did not reliably yield taxonomy-consistent outputs, underscoring the challenge of enforcing specific semantic perturbation categories in generative chemistry systems. This study provides a structured methodological foundation for future work investigating the boundary between linguistic register and semantic utility in text-conditioned molecular prediction.

Future directions.

Key extensions include: (1) investigation of the distributional compression phenomenon on additional enzymatic datasets, (2) expansion of multi-model validation across diverse LLM architectures and scales, (3) larger-scale evaluation with cross-validation, and (4) comparison with fine-tuned LLM models and additional traditional ML methods to further contextualize the practical significance of the observed effects.

Declarations

Competing Interests

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The complete codebase, generated datasets, and high-resolution figures are publicly available at: `https://github.com/apiprdt/AI-Hallucinations`.

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Use of AI and AI-assisted technologies

During the preparation of this work, the author used Llama-3.1-8B and Llama-3.3-70B for the generation of experimental hallucinations and molecular property prediction. These tools were also used for text refinement and grammar assistance. The author takes full responsibility for the content of the publication.

Ethics approval

Not applicable. This study does not involve human participants or animals.

Authors' contributions

Muhammad Afif Erdita: Conceptualization, Data Curation, Formal Analysis, Investigation, Methodology, Project Administration, Software, Validation, Visualization, Writing – Original Draft, Writing – Review & Editing.

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A Appendix

A.1 Prompt Templates

All prediction tasks used the following zero-shot prompt structure with temperature $= 0.0$, `max_tokens` $= 10$, and seed $= 42$:

```
Given this molecule (SMILES: {smiles})
[Additional information: {augmentation}]

On a scale of 0 to 1, what is the probability
that this molecule can {task_description}?
Respond with ONLY a number between 0 and 1.
Nothing else.
```

The augmentation line is omitted for C0. Task descriptions used for each benchmark:

- BBBP: "penetrate the blood-brain barrier"
- BACE: "inhibit the BACE-1 enzyme"
- Tox21: "exhibit toxicity in the nuclear receptor androgen receptor (NR-AR) assay"
- ESOL: "have low aqueous solubility, consistent with a logS value below the molecule's expected range"

Hallucination generation used temperature $= 0.9$, `max_tokens` $= 150$, with the following type-specific prompts:

C3 (Free Hallucination):

"You are a speculative chemist. Given SMILES: {smiles}. Generate a creative, scientifically-sounding description of this molecule's potential properties and biological mechanisms. Be creative and speculative. Write 2–3 sentences only."

C4a (Structural Augmentation):

"Given SMILES: {smiles}. Describe the structural features, functional groups, and atomic composition. Be specific and speculative."

C4b (Property Inversion):

"Given SMILES: {smiles}. Describe its solubility, lipophilicity, and membrane permeability with specific values."

C4c (Mechanism Fabrication):

"Given SMILES: {smiles}. Hypothesize biological targets and cellular mechanisms. Be specific about protein targets."

A.2 Control Condition Templates

C2 (Chemical Priming):

Template-generated text using scientific vocabulary: *"This compound exhibits {adj} structural coherence with {noun} interaction domains, demonstrating {adj} physicochemical characteristics under {noun} conditions."*

Adjective pool: moderate, intermediate, variable, standard, conventional, typical. Noun pool: hypothetical, theoretical, computational, structural, molecular, chemical.

Note: This template contains chemical vocabulary ("structural", "molecular", "physicochemical"), making C2 a *partial* stylistic control. See C2b for the stricter control.

C2b (Pure Gibberish):

Template-generated text using exclusively non-chemical vocabulary: *"This entity exhibits {adj} portfolio coherence with {noun} governance domains, demonstrating {adj} bureaucratic characteristics under {noun} conditions."*

Adjective pool: variable, standard, conventional, typical, moderate, intermediate. Noun pool: portfolio, transactional, governance, bureaucratic, administrative, regulatory.

Vocabulary audit confirmed 0% overlap with a curated chemical/biological term list.

A.3 Taxonomy Classifier Pseudocode

1. **SP Detection:** For each element in $\{F, Cl, Br, I, P, S, N\}$, check if associated keywords appear in the generated text. Cross-reference with `mol.GetAtoms()` to verify element absence. If a keyword matches an absent element, classify as SP.

2. **PI Detection:** Compute MolLogP via RDKit. If LogP > 3 and text contains "water soluble", "hydrophilic", or "highly soluble", classify as PI. If LogP < 1 and text contains "lipophilic", "hydrophobic", or "membrane permeable", classify as PI.

3. **MF Detection:** Keyword matching for 9 biological targets: ACE2, HER2, EGFR, VEGF, p53, BCL-2, COX-2, dopamine, serotonin.

4. **CC (Default):** If none of the above conditions trigger, classify as Contextual Confabulation.

A.4 Representative Hallucination Examples

- **SP Example (SMILES: c1ccccc1, Benzene):** "Contains a reactive chlorine atom that enhances metabolic stability." (Benzene contains no chlorine atoms.)

- **PI Example (SMILES: high-LogP compound):** "This highly water-soluble molecule readily dissolves in aqueous media." (Contradicts computed LogP > 3.)

- **MF Example (Diazepam):** "Acts as a potent inhibitor of the VEGF protein kinase receptor." (Diazepam is a GABA receptor modulator, not a VEGF inhibitor.)

- **CC Example:** "This molecule shows promising drug-like properties and may have therapeutic applications in treating various diseases." (Generic, non-falsifiable.)

A.5 API Failure Rates

Across the BBBP experiment (204 molecules $\times$ 9 conditions $= 1{,}836$ predictions), API failures (rate limits, parsing errors) that defaulted to $p = 0.5$ accounted for less than 2% of total predictions. The exact count is logged in the experiment checkpoint files.

A.6 Implementation Environment

Experiments were executed on a local system (Windows 11) using the Groq API for LLM inference. Total runtime for the pilot study was approximately 4.5 hours. Library versions: `rdkit-pypi` v2023.9.1 [21], `deepchem` v2.7.1 [22], `groq` Python SDK, `scipy` v1.11, `scikit-learn` v1.3.